Yingte Xu | PhD candidate · MPI-SP

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Profile

Yingte's research applies formal methods to study the programming theory for quantum computing, including semantics, verification techniques, and proof automation. The aim is a practical toolchain that can naturally encode and verify quantum algorithms and protocols.

Research Interests

Quantum Computing · Formal Methods · Programming Language Theory · LLM Architecture

Personal

Date of birth: 23 October 1999

Place of birth: Suzhou, Jiangsu

Nationality: P.R. China

Languages: Chinese (native), English, Japanese (N1)

Education & Employment

Max Planck Institute for Security and Privacy (MPI-SP)
Research Assistant *2024–present*

Ruhr University Bochum
PhD Student *2024–present*

Max Planck Institute for Security and Privacy (MPI-SP)
Remote Internship *2023–2024*

State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences
Internship *2022–2023*

Lanzhou University
Bachelor of Science (Physics) *2018–2022*

Suzhou High School *2015–2018*

Publications

[**LICS '26 (Accepted)**]: Gilles Barthe, Minbo Gao, Jam Kabeer Ali Khan, Matthijs Muis, Ivan Renison, Keiya Sakabe, Michael Walter, Yingte Xu, Tianshi Yu, Li Zhou. *Complete Relational Logic for Infinite-Dimensional Quantum Programs with Unbounded Assertions.*

[**CAV '25**]: Yingte Xu, Li Zhou, and Gilles Barthe. *D-Hammer: Efficient Equational Reasoning for Labelled Dirac Notation.* `arxiv.org/abs/2505.08633`

[**POPL '25**]: Yingte Xu, Gilles Barthe, and Li Zhou. *Automating Equational Proofs in Dirac Notation.* `arxiv.org/abs/2411.11617`

[**TOSEM '25**]: Yuan Feng, Li Zhou, Yingte Xu, and Xiaoquan Xu. *Refinement Calculus of Quantum Programs with Projective Assertions.* `arxiv.org/abs/2311.14215`

[**ASPLOS '23**]: Yuan Feng and Yingte Xu. *Verification of Nondeterministic Quantum Programs.* `arxiv.org/abs/2302.07973`

Awards

CHES 2025, GE Wars Organization Committee
Winner of CHES 2025 Deep Learning SCA Battle *2025*
pace-tl.gitbook.io/ches-challenge-2025

Ministry of Education of the People's Republic of China
National Scholarship *2019*

Projects

D-Hammer *2025*
A C++ tool that efficiently decides equations of labelled Dirac notations. github.com/LucianoXu/D-Hammer

DiracDec *2024-2025*
A prototype for proof automation of Dirac notation equations. Implemented in Mathematica, integrated with its computer algebra system for reals. github.com/LucianoXu/DiracImplementation

QRefine *2024*
A prototype IDE for quantum program development with program refinement support, implemented in Python. github.com/LucianoXu/QRefine

NQPV *2023*
Artifact for verification of nondeterministic quantum programs. github.com/LucianoXu/NQPV